REHAN KHAN

B.Tech CSE (AI/ML) Student | AI Systems | Agentic Orchestration & MCP-driven Ecosystems

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SUMMARY

Third-year (5th semester) B.Tech Computer Science Engineering (Artificial Intelligence and Machine Learning, AI/ML) student at Centurion University of Technology and Management with a CGPA of 8.87, and co-founder of NAVANTA.IO. Focused on engineering agentic AI orchestration, MCP (Model Context Protocol)-driven ecosystems and quant-optimized intelligence; actively exploring quantum computing, liquid neural networks (LNN), edge AI and IoT. Hands-on with Python, LLMs, RAG and AI agents. Seeking an AI/ML internship at a fast-moving startup.

EDUCATION

Centurion University of Technology and Management Sep 2024 - Aug 2028 (Expected)
Bachelor of Technology (B.Tech), Computer Science Engineering (AI/ML) *CGPA: 8.87*

Kendriya Vidyalaya (KV) Aug 2020 - Aug 2022
Senior Secondary, Science (PCMB) *Grade: 80.2%*

PROJECTS

RAG Docs Chat: Chat with Your Documents github.com/ReXhXan/rag-docs-chat
Tech: Python, FastAPI, ChromaDB, NumPy, sentence-transformers, OpenAI/Groq/Ollama, Streamlit, pytest, Docker

- Built a production-style Retrieval-Augmented Generation (RAG) pipeline with a vector database: page-aware PDF/DOCX/TXT/MD ingestion, recursive chunking with overlap, embeddings and top-k cosine search over ChromaDB or an in-memory NumPy store.
- Generated grounded answers with numbered citations (file, page, chunk) via pluggable Large Language Models (LLMs) and citation-enforcing prompt engineering; FastAPI REST API, Streamlit chat UI, retrieval evaluation (hit-rate@k, MRR), pytest suite and Docker Compose.

Smart NFC Business Card: AI-Powered Digital Card github.com/ReXhXan/smart-nfc-card
Tech: Python, FastAPI, SQLite, OpenAI/Groq/Ollama, qrcode, pytest, Docker

- Implemented NFC Forum NDEF URI record encoding with a QR fallback that opens a live profile card with one-tap vCard 3.0 download.
- Added an AI-written bio and an "Ask me" visitor chatbot grounded in IDF-ranked profile facts, plus SQLite tap analytics by source, device and day.

Self-Testing Code Assistant: Autonomous Coding Agent github.com/ReXhXan/self-testing-code-assistant
Tech: Python, pytest, FastAPI, OpenAI/Groq/Ollama, Docker

- Designed an AI agent loop with separate coder, tester and repairer LLM prompts that writes code and its own tests, then repairs the code from failures until tests pass or an iteration limit is hit.
- Ran tests in an isolated sandbox subprocess (python -I, clean env, timeout, CPU/memory rlimits) with AST syntax checks; exposed via CLI and a FastAPI REST API (POST /solve).

Jarvis: Personal AI Agent github.com/ReXhXan/jarvis-agent
Tech: Python, OpenAI/Groq/Ollama, SQLite, SpeechRecognition, pyttsx3, pytest, Docker

- Built a tool-calling AI agent on an MCP-style tool registry (name/description/inputSchema, JSON-schema validation) with calculator, weather, notes, reminders, file search and system info tools.
- Added persistent SQLite conversation memory, a provider-agnostic JSON tool-call protocol that works with small local Ollama models, and optional voice I/O.

TECHNICAL SKILLS

- **Languages & Frameworks:** Python, FastAPI (REST APIs), Pydantic, Streamlit, pytest
- **AI/Machine Learning:** Machine Learning, LLMs (OpenAI, Groq, Ollama APIs), RAG, embeddings (sentence-transformers), vector databases (ChromaDB), NumPy, AI Agents, tool calling, MCP-style tool registries, prompt engineering
- **Data & DevOps:** SQLite, Docker, Docker Compose, Git/GitHub
- **Exploring:** Quantum Computing, Liquid Neural Networks (LNN), Edge AI, Internet of Things (IoT)

EXPERIENCE

Co-Founder NAVANTA.IO

- Co-founded NAVANTA.IO while pursuing a full-time B.Tech in CSE (AI/ML).

ACHIEVEMENTS

- Silver Medal, Science Olympiad Foundation (SOF) Olympiad, Kendriya Vidyalaya.

ACTIVITIES

- **Member:** Coding Club, AI/ML Student Chapter, Design Club (UI/UX), Public Speaking Society, Centurion University.